

Point charge in a magnetic field

Arman Gabdullin

April 2026

1 Problem statement

A lightweight ring is attached to a fairly long rod, which can move freely along the rod without friction. A rubber cord is attached to the ring. At the other end of the cord, a particle of mass m and charge $q > 0$ is attached to which speed is imparted along the cord. Coefficient of stiffness of the cord is k . Initially, the length of the cord is equal to l_0 and the velocity of the particle v_0 is parallel to the cord.

Question 1: What is the trajectory of the particle? Find the parameters of its movement.

it is known that we could write the absolute value of the speed of particle as a function of time:

$$v(t) = v_0 \sqrt{c_1 + c_2 \cos(\alpha t)} \quad (1)$$

Where c_1, c_2 and α are constants.

Question 2: Find c_1, c_2 and α .

Remark: The values of the problem were selected so that $mv_0 < qBl_0$

